Ex. 9.4 Q30

$$\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^{3/2}}$$

$$\ln n \leq n^{p} \Rightarrow (\ln n)^2 \leq n^{2p}$$

$$\Rightarrow \frac{(\ln n)^2}{n^{3/2}} \leq \underline{\frac{n^{2p}}{n^{3/2}}} = \underline{\frac{1}{n^{3/2-2p}}}$$

$$3/2 - 2p > 1 \Rightarrow p < \frac{1}{4}$$

Choose $\underline{p = \frac{1}{5}}$ $\Rightarrow$ DCT

__Sol.__ NOTE THAT $\ln n \leq n^{1/5} \Rightarrow \ldots$